

MAT 426 - Topics in Linear Algebra – Final Review Exercises

Below are some questions over the material since Exam 2 to help you prepare for Final Exam on Friday, May 15. This list is not meant to be exhaustive and some topics may have been missed. Any topics that I have failed to include here are still considered reasonable exam material. Be sure to study all materials considered in class and in the homework.

1. State the definition of an inner product.
2. Verify that the following define an inner product.
 - (a) $\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 w_1 + u_2 v_2 w_2$ where $w_1, w_2 > 0$.
 - (b) Let $A, B \in M_{m \times n}(\mathbb{R})$. $\langle A, B \rangle = \text{trace}(B^T A)$ where $\text{trace}(A)$ is the sum of the diagonal entries of A .
 - (c) For polynomials $p(x), q(x)$ and given $x_0, x_1, \dots, x_n$, $\langle p(x), q(x) \rangle = \sum_{i=0}^n p(x_i)q(x_i)$
3. Let $f = x - 1$ and $g(x) = \frac{1}{2}x^2 - 2x + 1$. Consider the inner product $\langle f, g \rangle = \int_0^2 f(x)g(x) dx$. Compute the following.
 - (a) $\|g\|$
 - (b) $d(f, g)$
 - (c) $\langle f, g \rangle$
 - (d) Are f and g orthogonal?
4. Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$. Sketch the “unit circle” for the inner product $\langle \mathbf{u}, \mathbf{v} \rangle = 2u_1 v_1 + 7u_2 v_2$.
5. Use Gram-Schmidt to construct an orthogonal basis for the following set of linearly independent 2×2 matrices
$$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\}$$
relative to the inner product $\langle A, B \rangle = \text{trace}(B^T A)$.
6. Construct an orthogonal basis for $\mathcal{P}_2$ (the set of all polynomials of degree at most 2) with respect to the inner product
$$\langle f, g \rangle = \int_{-1}^1 f(x)g(x) dx$$
by applying the Gram-Schmidt process to the basis $\{1, 2x - 1, x^2\}$.
7. Prove the Cauchy-Schwarz Inequality
8.
 - (a) Let f and g be continuous function on $[0, \infty)$. Prove that $\langle f, g \rangle = \int_0^\infty f(x)g(x)e^{-x} dx$ is an inner product.
 - (b) Consider the basis $\{1, x, x^2, x^3, \dots\}$ for the set of all polynomials on the interval $(0, \infty)$. Use Gram-Schmidt to construct $\{L_0(x), L_1(x), L_2(x), L_3(x), \dots\}$, the set of orthogonal polynomials known as the Laguerre polynomials. These are typically normed so that $L_n(0) = 1$. Try to find through $L_3(x)$.
9. Find the Fourier series for $f(x) = x$ on $[-\pi, \pi]$.